

Subject Description Form

Subject Code	ITC3088E
Subject Title	3D Scan, Design, and Print in Fashion Design
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: SFT314FD 3D Printing for Fashion Design
Objectives	The subject introduces the basics of 3D CAD software and 3D printing (3DP) technologies for fashion design. It provides knowledge and hands-on experience of creating fashion related designs (e.g, accessories) using 3DP technology. It also explores 3DP applications in other fashion wearable products.
Intended Learning Outcomes	Upon completion of the subject, students will be able to learn the following. (a) Current 3DP technology advancement and its impacts in the fashion industry sectors (e.g., design, product development, retail, supply chain) (b) Current 3DP integrations in the fashion wearable products (c) Application of 3D CAD software and 3DP technology for fashion design and fashion wearable products (d) Understand the basic post-finishing techniques in 3DP fashion products (e) Defining and research target market for the appropriate 3DP fashion product integration (f) Using online platforms and sourcing for fashion design
Subject Synopsis/ Indicative Syllabus	1. 3DP technology advancement and its impacts in the fashion industry 2. 3DP integrations in the fashion wearable products 3. 3D scanning equipment, editing software, and reverse engineering 4. 3D CAD object formats 5. 3DP equipment 6. 3D CAD software for 3DP (accessories) 7. Post-finishing techniques for 3DP fashion products 8. Resources and costing for 3DP fashion products

Teaching/Learning Methodology	This subject is collaboratively developed and delivered by staff from the ITC and Industrial Centre (IC). It will be taught via the lecture and studio format. Students will be learning the 3D scanning techniques for different types of object and edit the scans digitally. Student will mainly focus on learning relevant 3D CAD software tools to render for 3D visualization and prototyping for 3DP. Using their knowledge from the lectures, students will be able to print their designs using 3D printers at IC. Students will be encouraged to explore the 3D scan, 3D CAD, 3DP, and post-finishing techniques for different fashion wearable products. There will be in-class discussions during lectures and presentations for student projects during studios.																																														
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Continuous Assessment</td><td>100%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Individual project (3D scan & 3D CAD)</td><td>35%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Group project (3D CAD & print)</td><td>65%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Individual Project assesses student’s knowledge and skills in using 3D scanning technology and 3D CAD tools (rendering and model) to develop fashion wearable product. In a collaborative manner, the Group Project assesses student’s knowledge and skills in 3D CAD in a collection and prototype using given 3DP technology. It also assesses student’s ability in defining and researching the target market and presented in a professional manner. The projects will be designed, developed, and assessed by ITC staff member. The materials submitted for all assessment must be the student’s own work. The submitted work may not be accepted for the purpose of	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	Continuous Assessment	100%	✓	✓	✓	✓	✓	✓	Individual project (3D scan & 3D CAD)	35%	✓	✓	✓				Group project (3D CAD & print)	65%	✓	✓	✓	✓	✓	✓	Total	100%						
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Total	100%																																														

	assessment if its authenticity is questionable. However, students could utilise various media and technologies such as GenAI to assist with design research in the preliminary stage. They are required to declare the use of GenAI in their work if applicable. Submitting GenAI-generated materials as students' own work or part of their work without declaration is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.	
Student Study Effort Expected	Class contact:	
	• Lecture (SFT)	12 Hrs.
	• Studio (SFT & IC)	27 Hrs.
	Other student study effort:	
	• Scan, design and print practice	28 Hrs.
	• Reading and researching	42 Hrs.
	Total student study effort	109 Hrs.
Reading List and References	<u>Publications</u> Lipson, H. and Kurman, M. (2013). <i>Fabricated: The new world of 3D printing</i>. John Wiley & Sons, Inc. Redwood, B. and Schoffer, F. (2017). <i>The 3D printing handbook: Technologies, design and applications</i>. 3D Hubs B.V. Bitonti, F. (2019). <i>3D printing design: Additive manufacturing and the materials revolution</i>. Bloomsbury Visual Arts. Sun, L. (2020). 3D printing technologies: Current applications, future trends, and challenges. In <i>Latest Material and Technological Developments for Activewear</i>. Elsevier Pub. Sun, L. & Zhao, L. (2017). <u>Envisioning the 3D printing era: A conceptual model for the fashion industry</u>. <i>Fashion and Textiles Journal</i>, 4(25), 1-16. Danaher, S. (2004), <i>Digital 3D design</i>. Thomson/Course Technology, Boston, MA. <u>Resources & Research</u> www.shapeways.com www.3dhubs.com www.thingiverse.com www.cults3d.com	

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